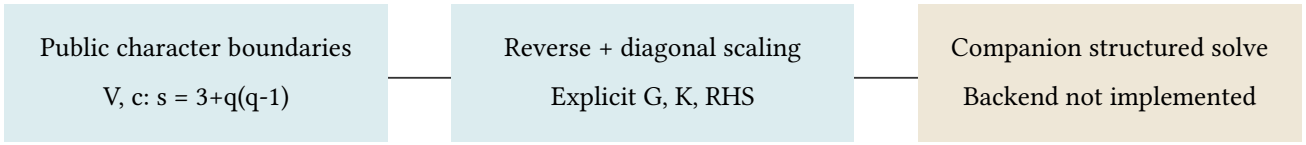
Direct cyclic companion reduction

Derived identities and unexecuted constructor · 2026-11-01

The existing weighted character Gram matrix has a cyclic displacement before its inverse-Pascal transform. An explicit reversal and diagonal scaling convert it to a standard companion displacement. The new constructor produces only public generators and RHS; no fast solver or finite validation has run.



No Pascal transform in the proposed generator constructor

Diagram of derived/proposed stages. Source equations: ../DERIVATION.md; prototype: ../generators.py. No measured-cost graph is applicable.

Boundary identity. Let (Tv)_k=v_(k+1 mod r), and let V collect the constant vector and both endpoint character vectors of every translated pattern. The same geometric weights as the current compiler telescope to B - gamma T B T^t = H = V diag(c) V^t, with s=3+q(q-1). Multiplying on the right by T gives gamma T B - B T = -H T.

Companion form. Let J reverse coordinates and D=diag(1,gamma,...,gamma^(r-1)). Put A=D^{-1} J B J, G=-D^{-1} J V diag(c), K=J T^t V. Then Z_(r,gamma) A - A Z_(r,1) = G K^t, where Z_(r,a) has subdiagonal ones and top-right a. The key prime-field identity is gamma^r=gamma, so D^{-1}(gamma T^t)D has exactly that wrap entry. The characteristic polynomials X^r-gamma and X^r-1 are coprime because gamma differs from 1.

No lost boundary data. If gamma T B=B T, iterate r times using T^r=I: (gamma-1)B=0, hence B=0. Thus the displacement is injective even though its characteristic polynomials have repeated roots in characteristic r.

Primal and dual transport. To solve Bx=b, solve Ay=D^{-1} Jb and set x=Jy. If vA=0 and vD^{-1} Jb=1, set w=vD^{-1} J; then wB=0 and wb=1. The separate public feature-exclusion test remains required. Here b_k=k is public.

Implementation scope. Constructor inputs are r,q,gamma,weights. It meters character powers, boundary vectors, diagonal powers, row scalings and RHS. It builds neither B nor a Pascal matrix, inverse or solution. O(rs) boundary data remain. Only syntax has been checked; coefficient identities and counter formulas still require the next frozen finite panel.

Primary-source connection. Bostan, Jeannerod, Mouilleron and Schost (2017) study companion displacement and fast structured systems. Their Theorem 6.7 has sufficient sampling condition |S| greater than 2n(n+1). For n=r, no S contained in F_r meets it. This limits use of that stated success guarantee; it does not prove the sampler fails. No extension-field source is substituted. Primary paper, SIAM J. Matrix Anal. Appl. 38(3),733–775.

Next gate. Freeze complete entrywise boundary/companion and primal/dual transport checks on the retained small and larger fixtures, with altered-wrap, scaling, generator and witness controls. Then implement the structured solve backend with explicit singular-system and base-field preconditioning behavior. The current full recovery instrument remains unchanged.